

ENVS-193DS-FINAL

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```
#package set up  
library(tidyverse)
```

```
-- Attaching core tidyverse packages ----- tidyverse 2.0.0 --  
v dplyr      1.2.1      v readr      2.2.0  
v forcats    1.0.1      v stringr    1.6.0  
v ggplot2     4.0.3      v tibble     3.3.1  
v lubridate  1.9.5      v tidyr      1.3.2  
v purrr       1.2.1  
-- Conflicts ----- tidyverse_conflicts() --  
x dplyr::filter() masks stats::filter()  
x dplyr::lag()     masks stats::lag()  
i Use the conflicted package (<http://conflicted.r-lib.org/>) to force all conflicts to become
```

```
library(janitor)
```

Attaching package: 'janitor'

The following objects are masked from 'package:stats':

chisq.test, fisher.test

```
library(readxl)  
library(DHARMA)
```

This is DHARMA 0.5.0. For overview type '?DHARMA'. For recent changes, type news(package = 'DHARMA').

Note that the default setting in simulateResiduals was changed to conditional simulations since version 0.5.0. To specify "unconditional" in simulateResiduals(). For more details, see ?simulateResiduals.

```
library(ggeffects)
nests <- read.csv("nest_data_final.csv")
walks <- read.csv("walks_data2.csv")
```

1

a)

In part 1 they used a binomial (logistical regression) test. In part 2, they used a chi-square test to check for a relationship.

b)

Part 1 : With a logistical regression we find that distance to water edge does predict the probability of American Avocet microhabitat use. Water edge proximity is an important variable American Avocet's use to determine where they live.

Part 2: We found that dependening on bird species, preference in habitat type also changes using a Chi-Square test. Habitat type preference is not consistent among varying bird species.

c)

One additional variable that could influence the probability of American Avocet microhabitat use is temperature. Temperature is a continuous variable (measured in either degrees celsius or fahrenheit) and could be indicative of microhabitat use along with water edge proximity as typically areas close to water have much more temperate climates, which could be a preference of the American Avocet.

2

a)

	height_cm	ant_species	case_control	scientific_name
1	236.0	C. mimosae	0	Crematogaster mimosae
2	213.5	C. mimosae	0	Crematogaster mimosae
3	259.5	C. mimosae	0	Crematogaster mimosae
4	114.5	C. mimosae	0	Crematogaster mimosae
5	109.0	C. nigriceps	0	Crematogaster nigriceps

6	353.0	C. mimosae	0	Crematogaster mimosae
7	168.5	C. mimosae	0	Crematogaster mimosae
8	93.0	C. nigriceps	0	Crematogaster nigriceps
9	121.5	C. mimosae	0	Crematogaster mimosae
10	181.5	C. nigriceps	1	Crematogaster nigriceps

b)

The 1s and 0s show whether or not a tree contains a nest (1) or is available for nesting (0)

c)

Tree height is a continuous variable. As tree height increases the probability of nest occurrence would increase because it provides more areas for birds to create nests in the area. (unable to access paper cannot cite this information, merely a guess)

Acacia Ant species is a categorical variable. The species of Acacia ant can fend off insects that harm bird offspring which increases the probability of nest occurrence. (abstract, paragraph 1).

d)

```
#create table
model_tabel <- tibble(
#model number column
  model_number = c(1,2),
  # predictor column
  predictors = c("height_cm + ant_species",
                 "height_cm * ant_species") ,
  #interaction term column
  interaction_term = c("No", "Yes"),
  #description column
  model_description = c("Contains tree height and ant species main effects", "Contains interaction")
)
```

e)

```
#create model 1
model0 <- lm(data = nests_clean,
  case_control ~ height_cm + ant_species
)
#create model 2
model1 <- lm(data = nests_clean,
  case_control ~ height_cm*ant_species)
```

f)

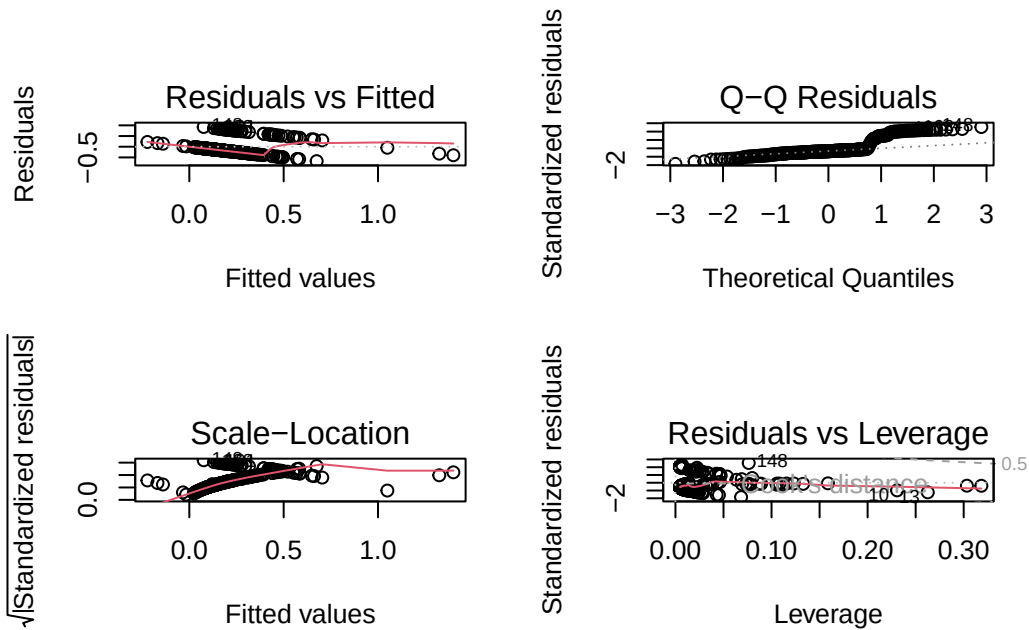
```
#run AIC between two models to determine strongest model
AIC(model0,model1)
```

	df	AIC
model0	5	263.7512
model1	7	248.1492

The best model as determined by Akaike's Information Criterion was the model predicting whether a tree was nested or not from tree height and ant species with an interaction term.

g)

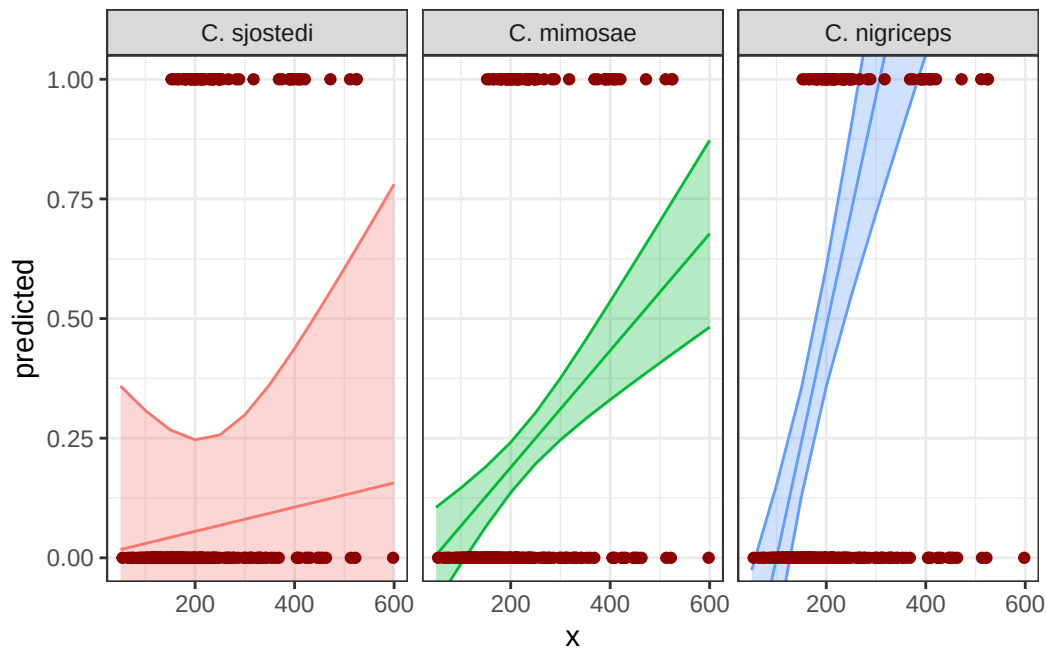
```
#plot diagnostics in 2x2 grid
par(mfrow = c(2,2))
#plot diagnostics
plot(model1)
```



h)

```
#predict data for model1
pred <- ggpredict(model1, terms = c("height_cm", "ant_species"))
#base ggplot, x = x, y = predicted value, color by group
ggplot(pred, aes (x = x, y = predicted, color = group,)) +
  #line
  geom_line() +
  #CI ribbon
  geom_ribbon(aes(ymin = conf.low, ymax = conf.high, fill = group), alpha = 0.3) +
  #googled how to keep predictions within 0-1 range due to binary restrictions
  coord_cartesian(ylim = c(0, 1)) +
  #jitter underlying data
  geom_jitter(data = nests_clean,
    # x = height of tree, y = case control (nest present or not)
    aes(x = height_cm, y = case_control),
    color = "red4",
    height = 0) +
  #wrap by ant species
  facet_wrap (~ group) +
```

```
theme_bw() +
theme(legend.position = "none")
```



i)

Figure 1 : Relationship between ant species and nested trees; nested trees represented as nested(1) and available (0) by red points, with ant species split and represented by green confidence interval, green confidence interval, and blue confidence interval. Data citation: Lujan, E., Nielsen, R., Short, Z., Wicks, S., Nderitu Watetu, W., Malingati Khasoha, L., Palmer, T. M., Goheen, J. R., & Alston, J. M. (2023). Data, code, and metadata for: Symbiotic acacia ants drive nesting behavior by birds in an African savanna [Data set]. Zenodo. <https://doi.org/10.5281/zenodo.8373322>

j)

```
#create new object, use ggpredict and model and set height to max height 600cm for prediction
max_height_pred <- ggpredict(model1, terms = c("height_cm [600]", "ant_species"))
max_height_pred
```

```
# Predicted values of case_control
```

```
ant_species: C. sjostedi
```

height_cm	Predicted	95% CI
600	0.16	-0.47, 0.78

```
ant_species: C. mimosae
```

height_cm	Predicted	95% CI
600	0.68	0.48, 0.87

```
ant_species: C. nigriceps
```

height_cm	Predicted	95% CI
600	2.41	1.70, 3.13

k)

The predicted probability of nest site occupancy at 600cm changes across ant species with the greatest probability lying with *C. nigriceps*, the second greatest with *C. mimosae*, and the lowest probability with *C. sjostedi*. With the figures in part h, we can see that *C. sjostedi* ants typically are found in shorter trees, and *C. nigriceps* typically being located in taller trees. Certain acacia ant species (*C. mimosae* and *C. nigriceps*) are more likely to be found at trees with bird nests, due to their ability to fend off insects that can harm bird offspring.

3

a)

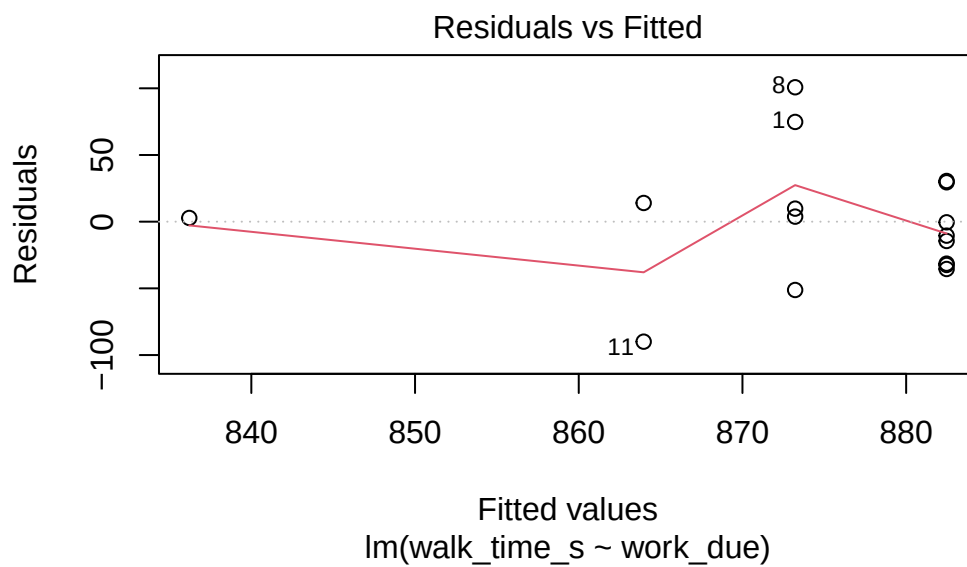
The visualizations are different because the affective visualization tells a story about the data whereas the visualizations in homework 2 show just the raw data, no story. Within the visualizations the only pattern I can see between the both of them is that they don't really vary, the data remains stable with few outliers in between. During week 9 workshop I hadn't created a visualization because I had submitted an extension for homework 3, so during the classmate feedback i bounced ideas off my classmates and they said that showing the route I would take was a very solid idea so I launched my visualization idea off of that interaction.

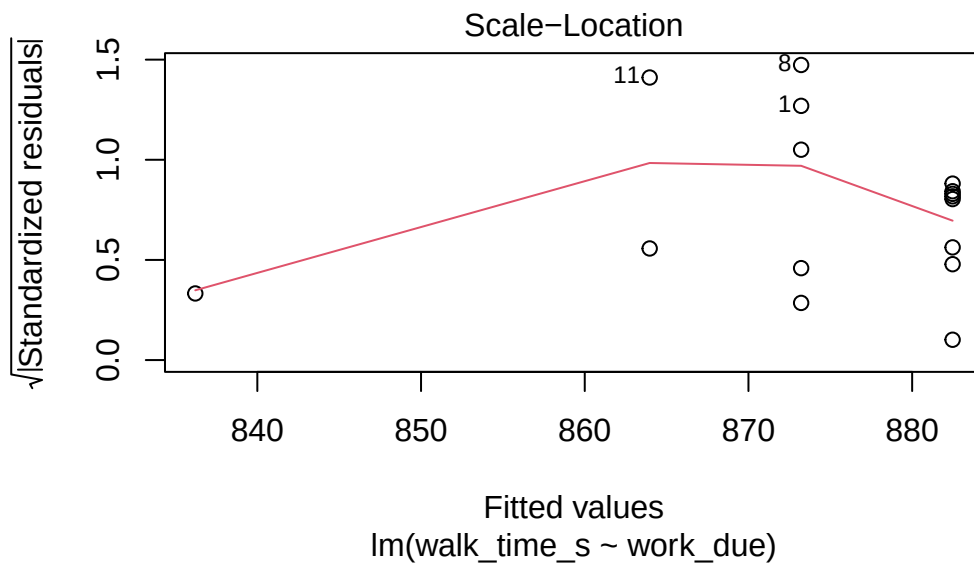
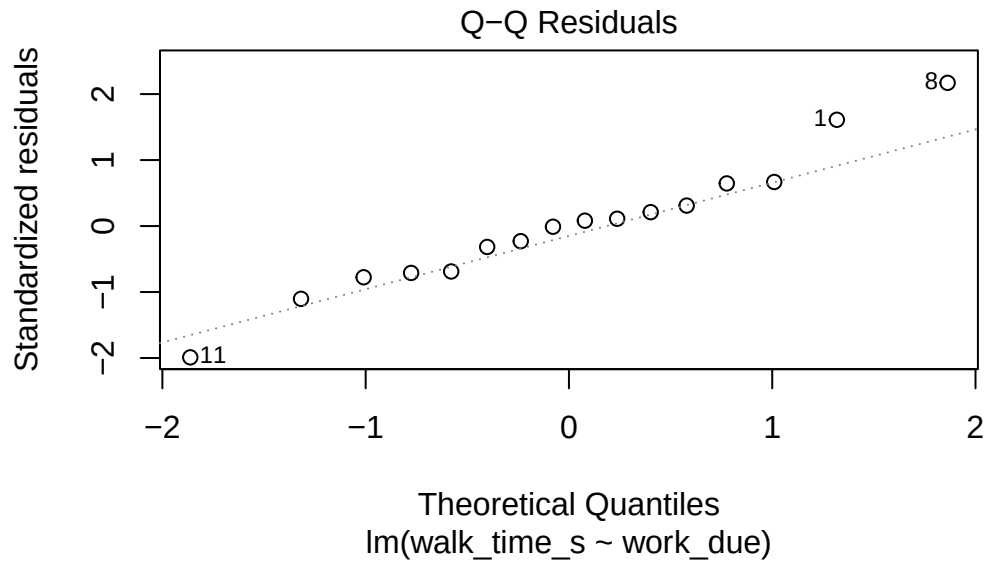
b)

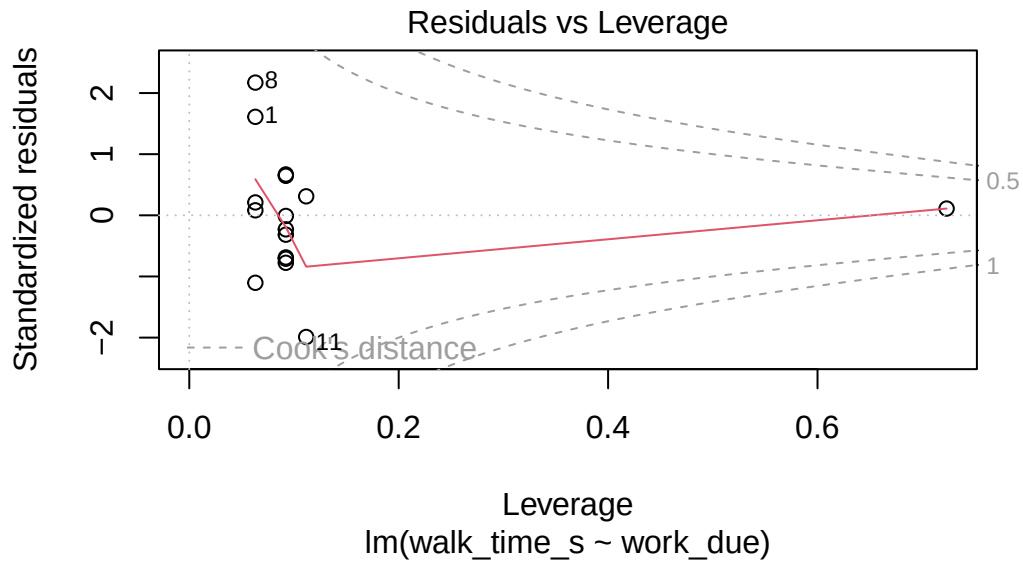
Based on my test of walk time and assignment workload, I believe a linear regression would be an appropriate test because workload was measured numerically and walk time is a continuous variable, which can be predicted with the linear regression.

c)

```
#walk model  
modelw <- lm(data = walks, walk_time_s ~ work_due)  
#plot model to check assumption  
plot(modelw)
```







```
#display model results
modelw
```

Call:

```
lm(formula = walk_time_s ~ work_due, data = walks)
```

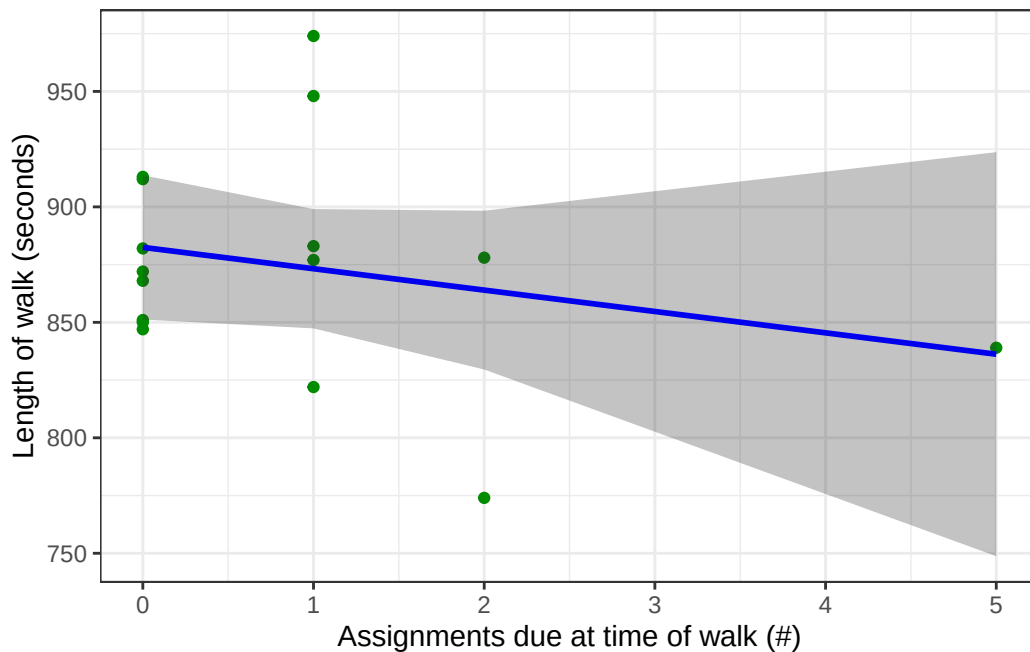
Coefficients:

(Intercept)	work_due
882.471	-9.252

d)

```
#predict walk data
walk_predictions <- ggpredict(modelw, terms = c("work_due"))
#base gg plot with predictions as base data
#x = x y = predicted values
ggplot(walk_predictions, aes (x = x, y = predicted)) +
  #geom point data = walks, x = work due, y = walk time in seconds
  geom_point(data = walks, aes(x = work_due, y = walk_time_s), color = "green4") +
  #confidence ribbon
```

```
geom_ribbon(aes(ymin = conf.low, ymax = conf.high), alpha = 0.3) +
geom_line(color = "blue2", linewidth = 1) +
theme_bw() +
#change labels
labs(x = "Assignments due at time of walk (#)",
     y = "Length of walk (seconds)")
```



e)

Figure 2 : Impact of workload on walk time; Blue line represents relationship between length of walk in seconds and work due, showing how with more work due the shorter the walk is. Represented in green points are the observed data with a 95% CI ribbon surrounding the data.

f)

The linear regression model for the relationship of work due and walk time over the course of the quarter ($n = 16$). The model shows that when 0 work is due, the predicted length of the walk is 882.47 seconds. For each assignment due this overall time is reduced by 9.25 seconds. This shows that there is a negative relationship between work due and length of walk.